Four cases, one comparison challenge

Learning intention

Compare named animal life cycles using similarities, differences and a valid counterexample.

All four animals lay eggs.

Does that make all four birds?

Your job is to test the claim with evidence.

Use the stage words carefully

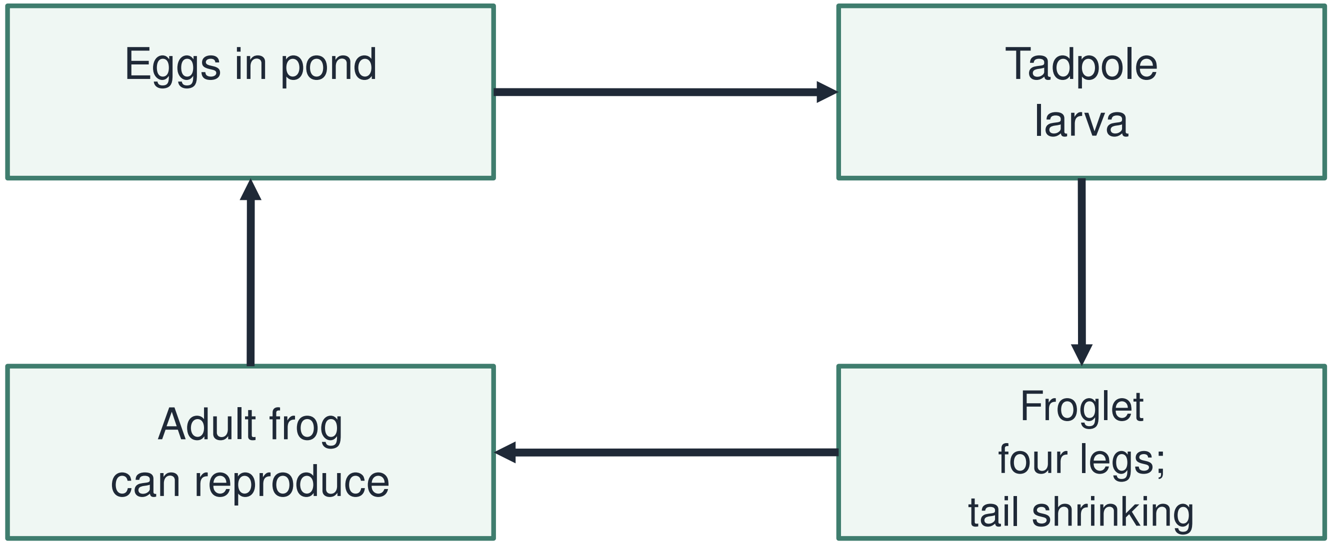
R1. What does an adult animal become able to do?

R2. Name an animal with a larva stage.

R3. Does one individual turn back into an egg?

I choose a feature, then compare

Common frog: major changes in body form

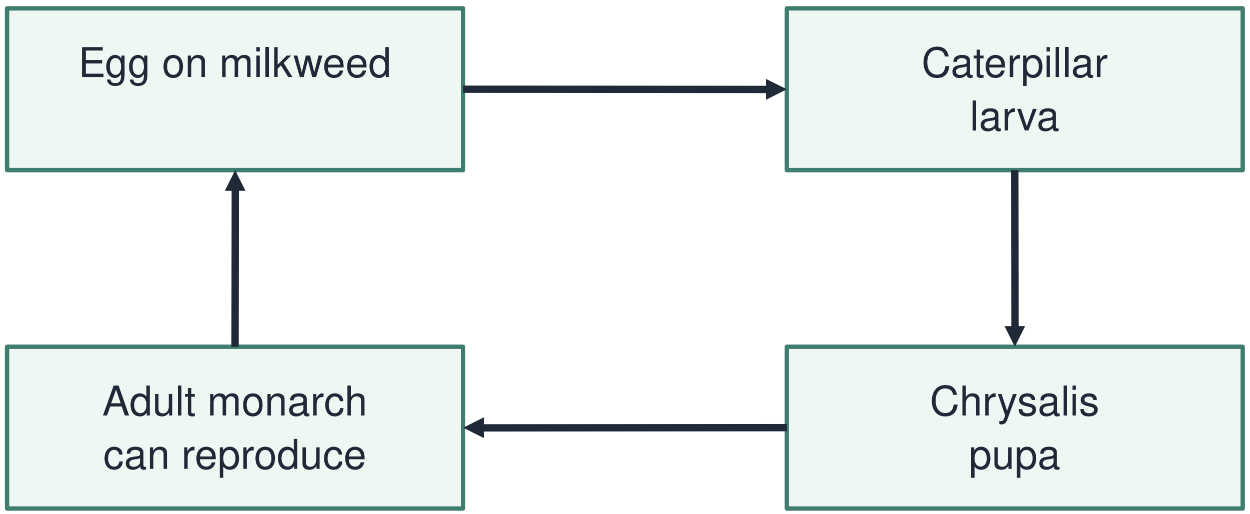


Adults produce the next generation of eggs.

I follow the common-frog stages. A tadpole and adult differ in body form. I look for a pupa; none is present.

I use the same feature for the monarch

Monarch: a different developmental route



A pupa is present here; it is not a stage in every insect.

Both routes include eggs and a larva. Only this monarch route has a pupa. A shared stage does not make the whole cycle identical.

I test an “all” claim

Compare one feature at a time

Same egg-laying feature; different animal groups.

Platypus and robin both lay eggs. Platypus young receive milk; robin chicks do not. The platypus disproves “all egg-layers are birds”.

Comparison bureau: shared-feature sort

Use cards F, M, P and R. W1. Group them by one feature, then regroup. Explain why the groups changed.

Feature	Question to use
Egg location	Where are eggs laid?
Pupa	Is this stage in the route?
Milk feeding	Do young receive milk?

Build a precise comparison

W2. Compare common frog with monarch.

Give one similarity and one difference.

Use “both”, “whereas” or a labelled pair.

The claim challenge

W3. “All egg-layers are birds.”

Select a counterexample and explain it.

Can four species describe every animal?

Which comparison uses evidence fairly?

Choose the statement that the cards support.

A Frog and monarch have identical life cycles because both lay eggs.

B Frog and monarch both have larvae, but only the monarch has a pupa.

C A platypus cannot be a mammal because it lays eggs.

Your independent investigation

Choose one response route.

Build an evidence-based comparison.

Test one broad claim with a named case.

Use the evidence

Use the same feature for both animals.

Name your species and evidence card.

Distinguish “these examples” from “all animals”.

Check before sharing

Does my explanation answer the question?

Have I named the evidence or process?

Have I kept the conclusion within its limits?

Compare and improve

Show a precise similarity or difference.

Check the named evidence.

Improve one broad or vague statement.

A question worth investigating

Suggest one next question about this lesson’s evidence.

Choose a question whose answer would improve our model.

Show the science you can explain

E1. Give a frog–monarch similarity.

E2. Give one difference in stages.

E3. Explain a counterexample to “all egg-layers are birds”.